

MYS LOPATKA, Russian Federation

WMO: 322130

Lat: **50.8734N**Lon: **156.6660E**Elev: **47**StdP: **100.76**Time Zone: **12.00 (E12)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.0	-8.2	-13.6	1.2	-8.0	-12.4	1.3	-7.5	27.4	-2.5	25.9	-2.5	11.0	320	1.094

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.4	12.9	12.0	12.2	11.5	11.7	11.1	12.3	12.7	11.7	12.0	11.2	11.5	6.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.1	8.8	12.5	11.5	8.5	11.9	11.1	8.2	11.4	35.1	12.7	33.6	12.2	32.4	11.6	17.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 23.4	20.7	18.7		DB	-10.5	15.4	1.0	1.2	-11.2	16.2	-11.8	16.9	-12.4	17.6	-13.1	18.5
(5)				WB	-11.2	14.2	0.9	1.6	-11.9	15.3	-12.4	16.3	-13.0	17.1	-13.6	18.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-3.8	-4.4	-3.3	-1.0	1.8	5.2	8.5	10.4	9.4	5.8	1.0	-2.1
	DBStd	5.46	2.13	2.49	1.99	1.46	1.21	1.33	1.35	1.12	1.21	1.86	2.22	2.33
	HDD10.0	2828	427	403	412	331	256	143	49	8	25	131	269	374
	HDD18.3	5839	685	637	671	581	514	393	305	246	268	389	519	633
	CDD10.0	31	0	0	0	0	0	0	3	21	7	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	9.0	10.4	11.1	10.6	9.7	8.4	7.4	6.7	6.6	7.3	9.3	10.0	10.8
Precipitation	PrecAvg	881	70	53	59	54	59	43	67	90	85	109	88	88
	PrecMax	1202	185	141	164	186	187	121	250	182	207	211	198	171
	PrecMin	498	13	2	12	5	13	6	7	19	12	46	22	31
	PrecStd	164	43	36	39	36	37	30	51	44	42	42	42	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	1.6	1.1	4.2	6.2	9.8	14.0	14.1	13.4	10.1	6.7	3.9
		MCWB	0.1	0.6	0.5	3.2	4.8	8.1	12.8	13.0	12.0	8.5	6.1	3.5
	2%	DB	0.5	0.6	0.2	1.6	4.8	8.4	12.4	13.1	12.3	9.3	6.0	3.0
		MCWB	-0.2	0.1	-0.3	0.8	3.8	7.3	11.5	12.2	11.3	8.0	5.5	2.4
	5%	DB	0.0	0.0	-0.1	1.1	4.1	7.6	11.2	12.5	11.8	8.7	5.1	2.1
		MCWB	-0.6	-0.4	-0.6	0.4	3.3	6.8	10.7	11.8	10.9	7.6	4.4	1.4
	10%	DB	-0.6	-0.7	-0.5	0.6	3.4	7.0	10.5	12.0	11.2	8.1	4.1	1.1
		MCWB	-1.3	-1.3	-1.1	0.0	2.9	6.5	10.1	11.5	10.4	7.1	3.3	0.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	0.8	0.6	3.4	5.3	8.6	13.1	13.6	12.5	9.1	6.5	3.7
		MCDB	0.8	0.9	0.9	4.3	5.7	9.1	13.8	13.8	13.0	9.5	6.6	3.8
	2%	WB	0.1	0.3	0.1	1.2	4.1	7.6	11.7	12.6	11.7	8.3	5.6	2.6
		MCDB	0.3	0.6	0.2	1.4	4.5	8.1	12.0	12.9	12.1	8.8	5.9	2.9
	5%	WB	-0.5	-0.3	-0.3	0.7	3.5	7.0	10.8	12.1	11.1	7.8	4.7	1.5
		MCDB	-0.1	0.0	-0.1	0.9	3.8	7.4	11.1	12.4	11.5	8.3	5.1	1.9
	10%	WB	-1.3	-1.2	-1.0	0.2	3.0	6.6	10.2	11.6	10.6	7.3	3.5	0.6
		MCDB	-0.7	-0.8	-0.6	0.5	3.4	6.9	10.5	11.9	11.0	7.8	3.9	1.0
Mean Daily Temperature Range	5% DB	MDBR	2.5	2.3	2.2	1.9	2.1	2.6	2.5	2.4	2.7	2.5	2.5	2.5
		MCDBR	2.9	2.7	2.2	2.4	3.2	3.5	3.8	3.2	3.4	2.9	3.5	3.1
		MCWBR	3.0	2.8	2.5	2.3	2.5	2.7	3.1	2.6	2.8	2.6	4.0	3.2
	5% WB	MCDDBR	2.7	2.6	2.3	2.3	3.1	3.1	3.5	2.9	3.3	2.8	3.5	2.5
		MCWBR	3.0	2.8	2.7	2.3	2.5	2.5	3.0	2.7	2.8	2.7	4.1	2.9
Clear-Sky Solar Irradiance	taub	0.332	0.358	0.407	0.458	0.484	0.458	0.454	0.430	0.377	0.392	0.369	0.337	
	taud	2.136	2.096	1.997	1.921	1.942	2.114	2.190	2.265	2.384	2.159	2.094	2.097	
	Ebn at Noon	648	738	769	775	776	803	800	800	809	691	587	566	
	Edh at Noon	79	107	142	172	176	150	137	121	97	100	82	71	
All-Sky Solar Radiation	RadAvg	0.85	1.51	2.60	3.69	4.23	4.70	4.31	3.81	3.15	1.92	0.98	0.66	
	RadStd	0.08	0.18	0.24	0.42	0.47	0.62	0.48	0.36	0.24	0.14	0.09	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page